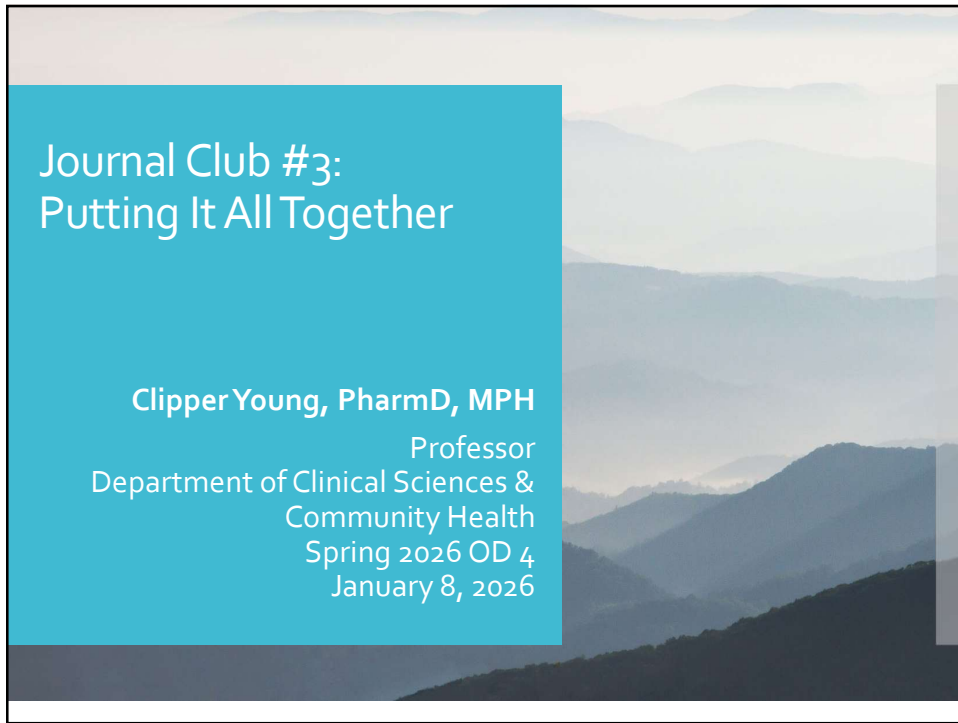
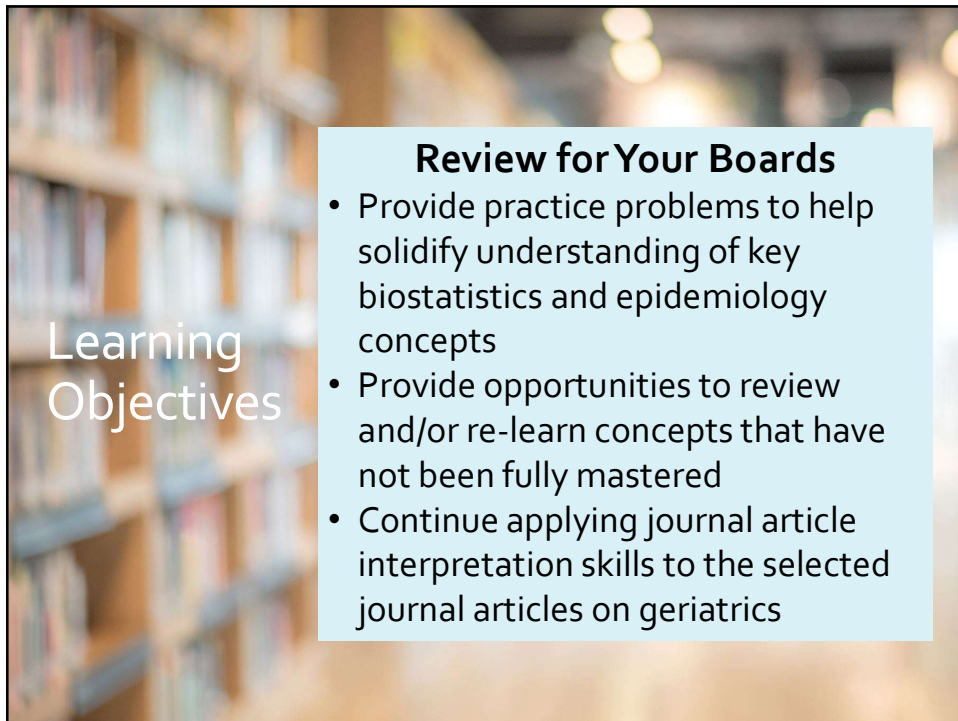




**Journal Club #3:
Putting It All Together**

Clipper Young, PharmD, MPH
Professor
Department of Clinical Sciences &
Community Health
Spring 2026 OD 4
January 8, 2026

1

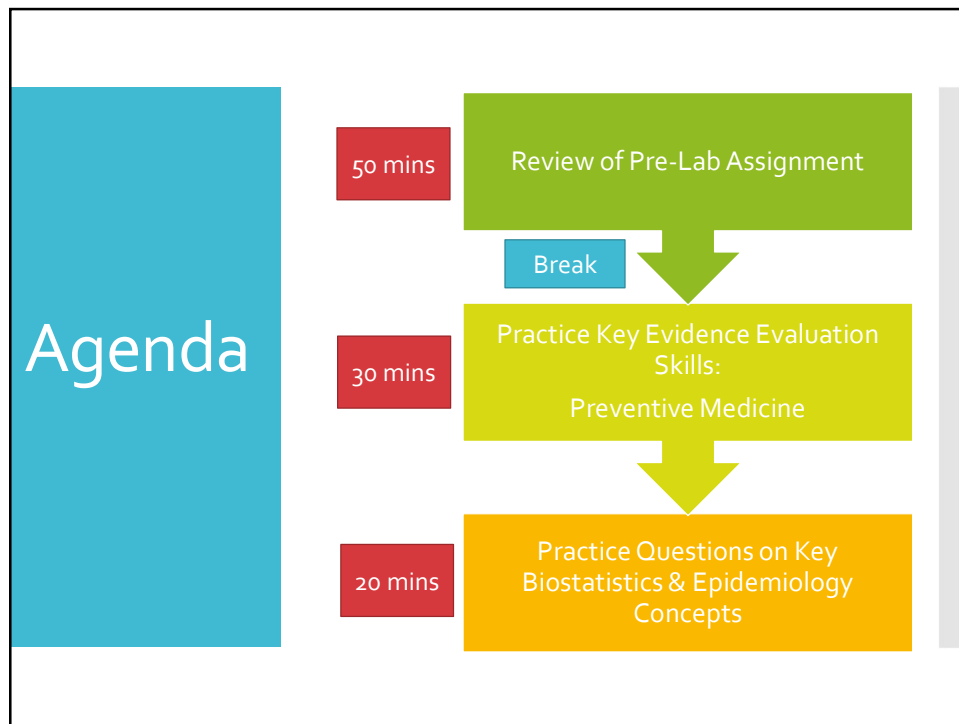


Learning Objectives

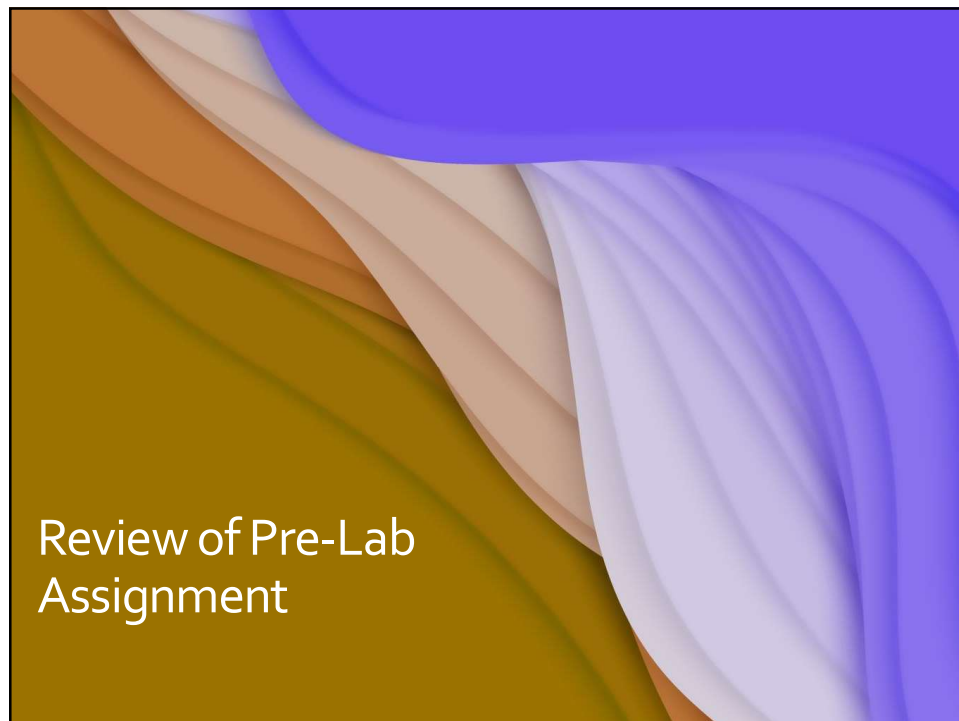
Review for Your Boards

- Provide practice problems to help solidify understanding of key biostatistics and epidemiology concepts
- Provide opportunities to review and/or re-learn concepts that have not been fully mastered
- Continue applying journal article interpretation skills to the selected journal articles on geriatrics

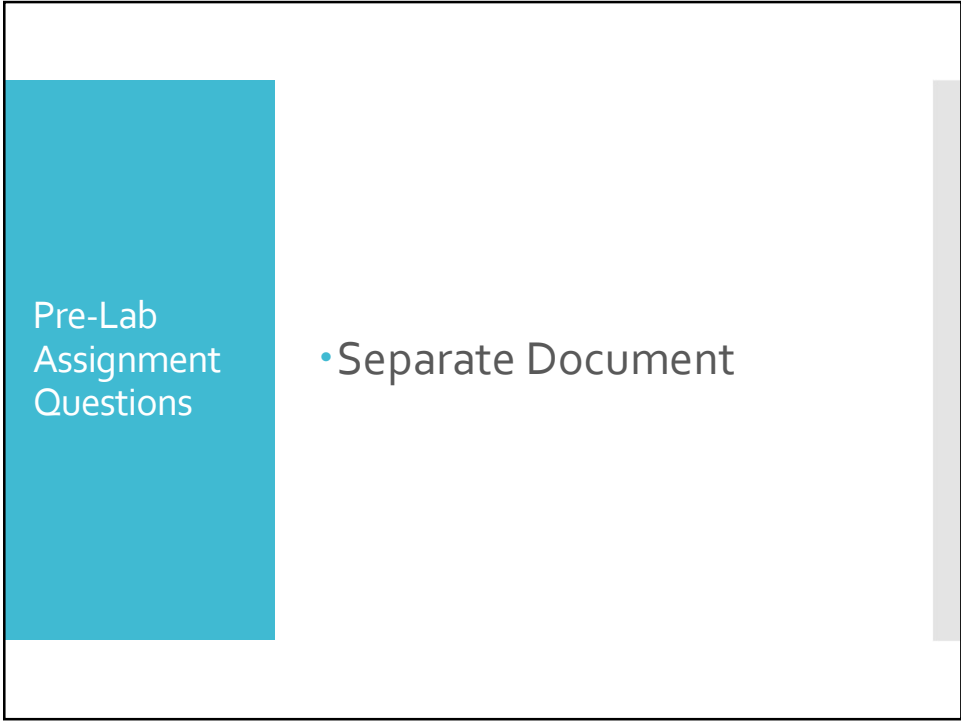
2



3



4

A presentation slide with a white background. On the left, there is a solid teal rectangle. To its right, the text "Pre-Lab Assignment Questions" is written in white. Further to the right, a bullet point is followed by the text "Separate Document".

Pre-Lab
Assignment
Questions

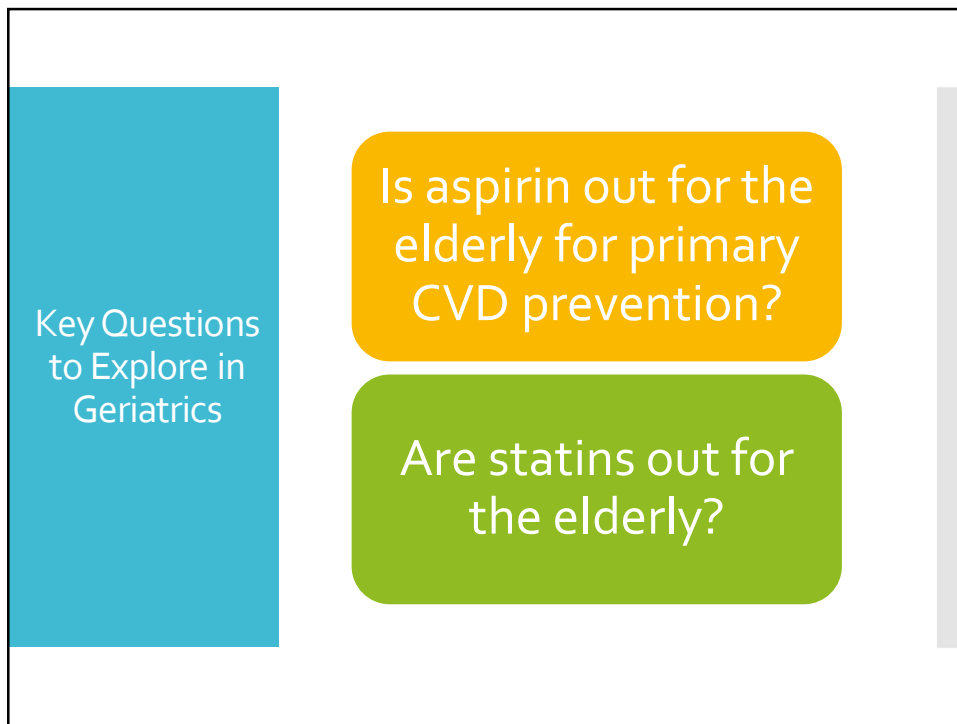
- Separate Document

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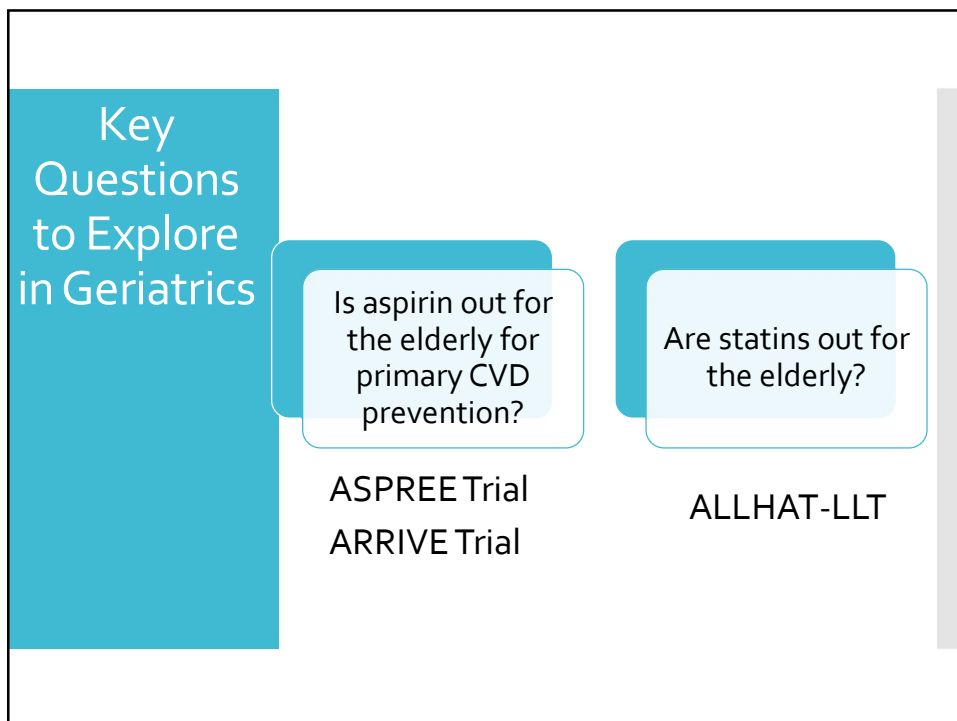
A presentation slide with a background of flowing, wavy lines in shades of blue, purple, and gold. The text "Practice Key Evidence Evaluation Skills: Preventive Medicine" is written in white in the lower-left corner.

Practice Key
Evidence Evaluation
Skills:
Preventive Medicine

6



7



8



9

Population	Agency & Year	Recommendation	Evidence Grade
Adults aged 50 to 59 years with a $\geq 10\%$ 10-year CVD risk	USPSTF 2016	Start low-dose aspirin for primary prevention	B
Adults aged 60 to 69 years with a $\geq 10\%$ 10-year CVD risk	USPSTF 2016	Start low-dose aspirin for primary prevention should be an individual decision	C
Adults aged 70 years or older	USPSTF 2016	Insufficient evidence	I

Aspirin Use to Prevent Cardiovascular Disease and Colorectal Cancer: Preventive Medicine
<https://www.uspreventiveservicestaskforce.org/uspstf/recommendation/aspirin-to-prevent-cardiovascular-disease-and-cancer>

10

ORIGINAL ARTICLE

Effect of Aspirin on Cardiovascular Events and Bleeding in the Healthy Elderly

J.J. McNeil, R. Wolfe, R.L. Woods, A.M. Tonkin, G.A. Donnan, M.R. Nelson, C.M. Reid, J.E. Lockery, B. Kirpach, E. Storey, R.C. Shah, J.D. Williamson, K.L. Margolis, M.E. Ernst, W.P. Abhayaratna, N. Stocks, S.M. Fitzgerald, S.G. Orchard, R.E. Trevaks, L.J. Beilin, C.I. Johnston, J. Ryan, B. Radziszewska, M. Jelinek, M. Malik, C.B. Eaton, D. Brauer, G. Cloud, E.M. Wood, S.E. Mahady, S. Satterfield,* R. Grimm, and A.M. Murray, for the ASPREE Investigator Group†

N ENGL J MED 379:16 NEJM.ORG OCTOBER 18, 2018

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Table 1. Demographic Characteristics, Cardiovascular Risk Factors, and Treatment of the Participants at Randomization.*

Variable	Aspirin (N=9525)	Placebo (N=9589)
	no. (%)	
Male sex	4152 (44)	4179 (44)
Age ≥74 yr	4806 (50)	4766 (50)
Black race†	451 (5)	450 (5)
Obese‡	2820 (30)	2857 (30)
Smoking		
Current	352 (4)	383 (4)
Former	3909 (41)	3890 (41)
Never	5264 (55)	5316 (55)
Diabetes§	1027 (11)	1030 (11)
Hypertension¶	7065 (74)	7148 (75)
Dyslipidemia	6159 (65)	6308 (66)
Chronic kidney disease**	2456 (26)	2464 (26)
Number of cardiovascular risk factors††		
0 or 1	2935 (31)	2885 (30)
2	3968 (42)	4049 (42)
3 or 4	2622 (28)	2655 (28)
Previous regular aspirin use‡‡	1053 (11)	1041 (11)
Statin use at trial entry§§	3244 (34)	3226 (34)
Use of nonsteroidal antiinflammatory drug at trial entry	1371 (14)	1342 (14)
Use of H ₂ -receptor blocker at trial entry	189 (2)	183 (2)
Use of proton-pump inhibitor at trial entry	2340 (25)	2374 (25)

* For all variables, odds ratios were between 0.67 and 1.50 (with an odds ratio of 1.50 as the prespecified limit for performing an adjusted analysis of end points) and there were no significant ($P>0.05$) differences between the two trial groups.

† Data on race were reported by the participants.

‡ Obese was defined as a body-mass index (the weight in kilograms divided by the square of the height in meters) of 30 or higher. Data were available for 19,035 participants; data on height, weight, or both were missing for 40 participants in the aspirin group and 39 in the placebo group.

§ The presence of diabetes was based on participants' report of diabetes mellitus or a fasting glucose level of at least 126 mg per deciliter (≥7 mmol per liter) or receipt of treatment for diabetes.

¶ Hypertension was defined as treatment for high blood pressure or a blood pressure of more than 140/90 mm Hg at trial entry.

|| Dyslipidemia was defined as the receipt of cholesterol-lowering medication or as a serum cholesterol level of at least 212 mg per deciliter (≥5.5 mmol per liter) in Australia and at least 240 mg per deciliter (≥6.2 mmol per liter) in the United States or as a low-density lipoprotein level of more than 160 mg per deciliter (>4.1 mmol per liter).

** Chronic kidney disease was defined as an estimated glomerular filtration rate of less than 60 ml per minute per 1.73 m² or a ratio of albumin (in milligrams per liter) to creatinine (in millimoles per liter) in urine of 3 or more.

†† Cardiovascular risk factors were diabetes, hypertension, dyslipidemia, and current or former smoking.

‡‡ Data were available for 19,112 participants; data were missing for 2 participants in the placebo group.

§§ Data for statin use included 483 participants (247 in the aspirin group and 236 in the placebo group) who were taking nonstatin lipid-lowering therapies.

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Table 2. Cardiovascular Events.*

End Point	Overall (N=19,114)	Aspirin (N=9525)	Placebo (N=9589)		Hazard Ratio (95% CI)	
	no. of participants with event	no. of participants with event	rate per 1000 person-yr	no. of participants with event		rate per 1000 person-yr
Cardiovascular disease†	922	448	10.7	474	11.3	0.95 (0.83–1.08)
Major adverse cardiovascular event‡	701	329	7.8	372	8.8	0.89 (0.77–1.03)
Fatal cardiovascular disease§	159	78	1.8	81	1.9	0.97 (0.71–1.33)
Hospitalization for heart failure	171	88	2.1	83	1.9	1.07 (0.79–1.44)
Fatal or nonfatal myocardial infarction	355	171	4.0	184	4.3	0.93 (0.76–1.15)
Fatal or nonfatal ischemic stroke¶	315	148	3.5	167	3.9	0.89 (0.71–1.11)

* Because some end points were composites, a participant who had events for more than one component of the composite (e.g., stroke and then acute myocardial infarction) would contribute only the first event that occurred to the composite end point but would contribute an event to the separate analyses of each component. Hence, summation of the number of events for separate components of a composite end point does not equate to the number of events for the composite end point. The confidence intervals shown in this table have not been adjusted for multiple comparisons, and inferences drawn from them may not be reproducible.

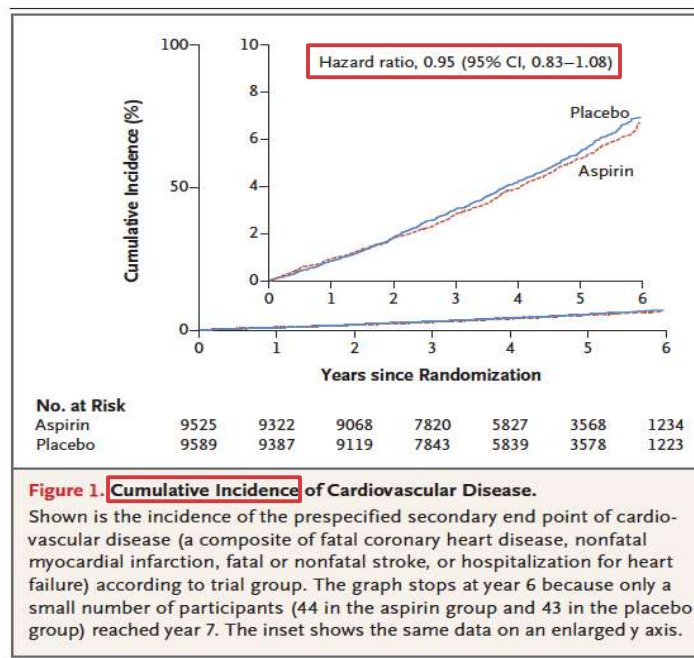
† Cardiovascular disease, a prespecified secondary end point, was a composite of fatal coronary heart disease (death from myocardial infarction, sudden cardiac death, or any other death in which the underlying cause was considered to be coronary heart disease), nonfatal myocardial infarction, fatal or nonfatal stroke (including hemorrhagic stroke), or hospitalization for heart failure.

‡ Major adverse cardiovascular events, a nonprespecified end point, was a composite of fatal coronary heart disease (excluding death from heart failure), nonfatal myocardial infarction, or fatal or nonfatal ischemic stroke.

§ Fatal cardiovascular disease was defined as any death from stroke (including hemorrhagic stroke) or coronary heart disease.

¶ Data for ischemic stroke included cases that were adjudicated as ischemic stroke, cases for which stroke type was uncertain after adjudication, and cases of ischemic stroke with hemorrhagic transformation.

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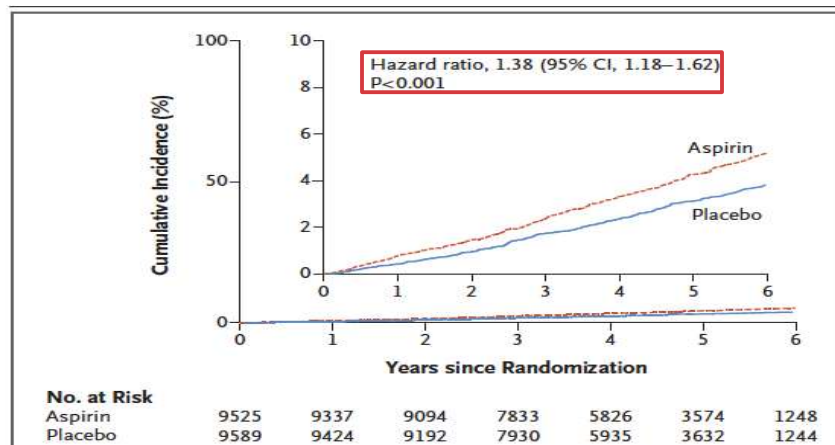


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Table 3. Major Hemorrhagic Events.*

End Point	Overall (N=19,114)	Aspirin (N=9525)		Placebo (N=9589)		Hazard Ratio (95% CI)	P Value
	no. of participants with event	no. of participants with event	rate per 1000 person-yr	no. of participants with event	rate per 1000 person-yr		
Major hemorrhage†	626	361	8.6	265	6.2	1.38 (1.18–1.62)	<0.001
Intracranial bleeding							
Any	179	107	2.5	72	1.7	1.50 (1.11–2.02)	—
Hemorrhagic stroke	77	43	1.0	34	0.8	1.27 (0.81–2.00)	—
Subdural or extradural hemorrhage	61	39	0.9	22	0.5	1.79 (1.06–3.02)	—
Subarachnoid hemorrhage‡	32	18	0.4	14	0.3	1.30 (0.64–2.60)	—
Extracranial bleeding							
Upper gastrointestinal bleeding	137	89	2.1	48	1.1	1.87 (1.32–2.66)	—
Lower gastrointestinal bleeding	127	73	1.7	54	1.3	1.36 (0.96–1.94)	—
Bleeding at another site§	189	101	2.4	88	2.1	1.16 (0.87–1.54)	—
Fatal bleeding							
Fatal major hemorrhage¶	52	28	0.7	24	0.6	1.18 (0.68–2.03)	—
Fatal hemorrhagic stroke	26	13	0.3	13	0.3	1.01 (0.47–2.17)	—

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**Figure 2. Cumulative Incidence of Major Hemorrhage.**

Shown is the incidence of the prespecified secondary end point of major hemorrhage (a composite of hemorrhagic stroke, symptomatic intracranial bleeding, or extracranial bleeding that led to transfusion, hospitalization, prolongation of hospitalization, surgery, or death) according to trial group. The graph stops at year 6 because only a small number of participants (44 in the aspirin group and 43 in the placebo group) reached year 7. The inset shows the same data on an enlarged y axis.

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ASPREE Trial

Sample Size & Study Population

- Enrolled 19,114 persons without cardiovascular disease, dementia, or disability to receive aspirin 100 mg daily or placebo
- Of those enrolled, 9,542 were between 65 and 73 years of age and 9,572 were > 74 years of age

Results

(4.7 years median follow-up period)

- Rate of cardiovascular disease was
- 10.7 events per 1000 person-years in the aspirin group
- 11.3 events per 1000 person-years in the placebo group
- HR 0.95; 95% confidence interval [CI], 0.83 to 1.08

Adverse Effect

- Rate of major hemorrhage was
- 8.6 events per 1000 person-years in the aspirin group
- 6.2 events per 1000 person-years in the placebo group
- HR 1.38; 95% CI, 1.18 to 1.62; $P < 0.001$

McNeil JJ, Wolfe R, Woods RL, et al. Engl J Med . 2018;379(16):1509–1518.

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ASPREE Trial

CONCLUSIONS

The use of low-dose aspirin as a primary prevention strategy in older adults resulted in a significantly higher risk of major hemorrhage and did not result in a significantly lower risk of cardiovascular disease than placebo. (Funded by the National Institute on Aging and others; ASPREE ClinicalTrials.gov number, NCT01038583.)

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Question:

Is aspirin beneficial when used for reducing *all-cause mortality* in healthy older adults?

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ORIGINAL ARTICLE

Effect of Aspirin on All-Cause Mortality in the Healthy Elderly

J.J. McNeil, M.R. Nelson, R.L. Woods, J.E. Lockery, R. Wolfe, C.M. Reid, B. Kirpach, R.C. Shah, D.G. Ives, E. Storey, J. Ryan, A.M. Tonkin, A.B. Newman, J.D. Williamson, K.L. Margolis, M.E. Ernst, W.P. Abhayaratna, N. Stocks, S.M. Fitzgerald, S.G. Orchard, R.E. Trevaks, L.J. Beilin, G.A. Donnan, P. Gibbs, C.I. Johnston, B. Radziszewska, R. Grimm, and A.M. Murray, for the ASPREE Investigator Group*

N ENGL J MED 379:16 NEJM.ORG OCTOBER 18, 2018

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Table 1. Mortality According to the Underlying Cause of Death.*

Cause of Death	Overall (N=19,114)	Aspirin (N=9525)	Placebo (N=9589)	Hazard Ratio (95% CI)
	no. of deaths	no. of deaths (%)	no. of deaths (%)	
Any	1052	558 (5.9)	494 (5.2)	1.14 (1.01–1.29)
Cancer†	522	295 (3.1)	227 (2.3)	1.31 (1.10–1.56)
Cardiovascular disease, including ischemic stroke‡	203	91 (1.0)	112 (1.2)	0.82 (0.62–1.08)
Major hemorrhage, including hemorrhagic stroke§	53	28 (0.3)	25 (0.3)	1.13 (0.66–1.94)
Other¶	262	140 (1.5)	122 (1.3)	1.16 (0.91–1.48)
Insufficient information	12	4 (<0.1)	8 (0.1)	—

* The confidence intervals shown in this table have not been adjusted for multiple comparisons, and inferences drawn from them may not be reproducible.

† Data are shown for deaths that were related to primary or metastatic cancer.

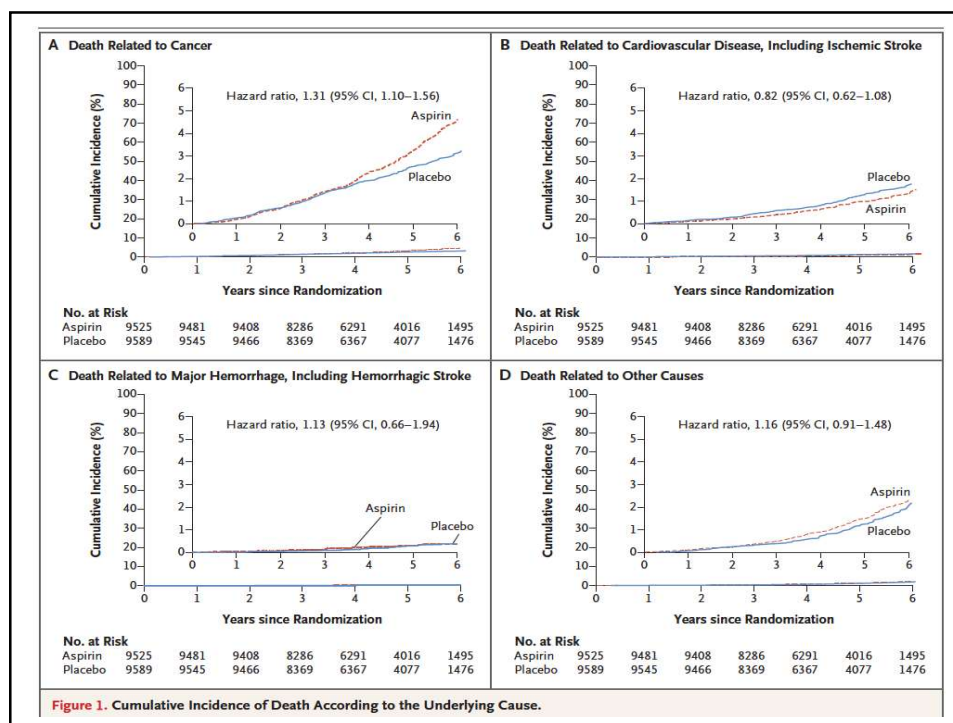
‡ Cardiovascular disease was defined as any ischemic event (myocardial infarction, other coronary heart disease, sudden cardiac death, or ischemic stroke).

§ Major hemorrhage was defined as any hemorrhagic event (hemorrhagic stroke, symptomatic intracranial bleeding, or major gastrointestinal bleeding or other extracranial bleeding).²

¶ Data are shown for deaths that were related to causes that are not represented in the other categories (e.g., sepsis, chronic lung disease, dementia, or heart failure), except for deaths for which insufficient information was available to adjudicate an underlying cause.

|| Data are shown for deaths for which insufficient information was available to adjudicate an underlying cause, even after linkage with the National Death Index. The hazard ratio was not determined because of the low numbers of deaths.

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Table 2. Cancer-Related Mortality According to Type of Cancer.*

Type of Cancer	Aspirin (N=9525)		Placebo (N=9589)		Hazard Ratio (95% CI)
	<i>no. of deaths</i>	<i>rate per 1000 person-yr</i>	<i>no. of deaths</i>	<i>rate per 1000 person-yr</i>	
Any	295	6.7	227	5.1	1.31 (1.10–1.56)
Lung	55	1.3	51	1.2	1.09 (0.74–1.59)
Colorectal	35	0.8	20	0.5	1.77 (1.02–3.06)
Pancreatic	29	0.7	21	0.5	1.40 (0.80–2.45)
Hematologic	24	0.6	19	0.4	1.27 (0.70–2.33)
Prostate	18	0.4	17	0.4	1.07 (0.55–2.08)
Ovarian or endometrial	15	0.3	13	0.3	1.16 (0.55–2.45)
Breast	15	0.3	7	0.2	2.16 (0.88–5.31)
Brain	13	0.3	9	0.2	1.46 (0.62–3.41)
Melanoma	12	0.3	8	0.2	1.51 (0.62–3.70)
Bladder	8	0.2	9	0.2	—
Kidney	9	0.2	4	0.1	—
Gallbladder or bile duct	7	0.2	6	0.1	—
Stomach	9	0.2	3	0.1	—
Liver	4	0.1	0	0	—
Other	28	0.6	27	0.6	1.05 (0.62–1.78)
Unknown primary	12	0.3	7	0.2	1.73 (0.68–4.39)
Undetermined	2	0.05	6	0.1	—

* Some hazard ratios were not determined because of the low numbers of deaths. The confidence intervals shown in this table have not been adjusted for multiple comparisons, and inferences drawn from them may not be reproducible.

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ASPREE Trial

CONCLUSIONS

Higher all-cause mortality was observed among apparently healthy older adults who received daily aspirin than among those who received placebo and was attributed primarily to cancer-related death. In the context of previous studies, this result was unexpected and should be interpreted with caution. (Funded by the National Institute on Aging and others; ASPREE ClinicalTrials.gov number, NCT01038583.)

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ARRIVE Trial

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Use of aspirin to reduce risk of initial vascular events in patients at moderate risk of cardiovascular disease (ARRIVE): a randomised, double-blind, placebo-controlled trial

J Michael Gaziano, Carlos Brotons, Rosa Coppolecchia, Claudio Cricelli, Harald Darius, Philip B Gorelick, George Howard, Thomas A Pearson, Peter M Rothwell, Luis Miguel Ruilope, Michal Tendera, Gianni Tognoni; the ARRIVE Executive Committee

Lancet 2018; 392: 1036–46

Published Online

August 26, 2018

[http://dx.doi.org/10.1016/](http://dx.doi.org/10.1016/S0140-6736(18)31924-X)

S0140-6736(18)31924-X

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	Aspirin (n=6270)	Placebo (n=6276)
Mean age, years	63.9 (7.1)	63.9 (7.1)
Sex		
Female	1851 (29.5%)	1857 (29.6%)
Male	4419 (70.5%)	4419 (70.4%)
Race		
White	6133 (97.8%)	6146 (97.9%)
Other	137 (2.2%)	130 (2.1%)
Current cigarette smoker*	1808 (28.8%)	1786 (28.5%)
Median weight, kg	82.0 (35–163)	82.0 (43–177)
Mean body-mass index	28.3 (4.3)	28.5 (4.3)
High total cholesterol†	3647 (58.2%)	3657 (58.3%)
High LDL‡	2775 (44.3%)	2869 (45.7%)
Low HDL§	857 (13.7%)	875 (13.9%)
High systolic blood pressure¶	3916 (62.5%)	3950 (62.9%)
Median systolic blood pressure	145.0 (80–199)	145.0 (95–215)
Taking anti-hypertensive medications	4038 (64.4%)	4097 (65.3%)
Country		
Germany	1525 (24.3%)	1525 (24.3%)
Italy	164 (2.6%)	171 (2.7%)
Ireland	54 (0.9%)	58 (0.9%)
Poland	1550 (24.7%)	1545 (24.6%)
Spain	212 (3.4%)	200 (3.2%)
UK	2518 (40.2%)	2510 (40.0%)
USA	247 (3.9%)	267 (4.3%)
Mean Framingham 10-year coronary heart disease risk score	13.9% (6.4)	14.1% (6.4)
Mean estimate ACC/AHA 10-year ASCVD risk score at baseline	17.3% (9.8)	17.4% (9.7)

Data are mean (SD), n (%), or median (range). ACC=American College of Cardiology; AHA=American Heart Association; ASCVD=atherosclerotic cardiovascular disease.

*Any cigarette smoking in the past 12 months or continuing smoker at randomisation. †Defined as concentrations greater than 200 mg/dL (5.180 mmol/L) in men, and greater than 240 mg/dL (6.216 mmol/L) in women. ‡Defined as concentrations greater than 130 mg/dL (3.367 mmol/L) in men, and greater than 160 mg/dL (4.144 mmol/L) in women. §HDL concentrations less than 40 mg/dL (1.036 mmol/L) at screening for both sexes. ¶Systolic blood pressure greater than 140 mm Hg at screening.

Table 1: Baseline characteristics of the intention-to-treat population

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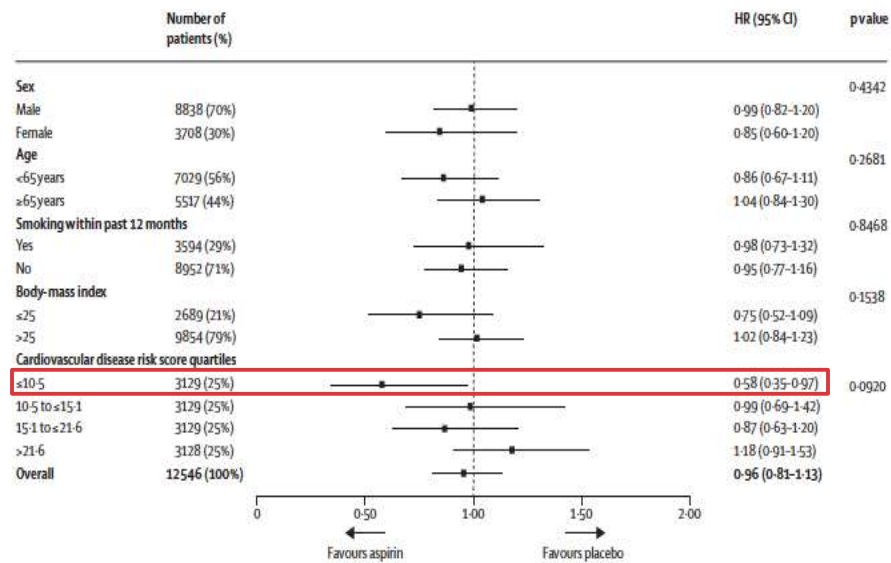
	Number of events in the intention-to-treat population			Number of events in the per-protocol population		
	Aspirin (n=6270)	Placebo (n=6276)	Hazard ratio (95% CI); p value	Aspirin (n=3790)	Placebo (n=3912)	Hazard ratio (95% CI); p value
Myocardial infarction, stroke, cardiovascular death, unstable angina, or transient ischaemic attack	269 (4.29%)	281 (4.48%)	0.96 (0.81-1.13); p=0.6038	129 (3.40%)	164 (4.19%)	0.81 (0.64-1.02); p=0.0756
Myocardial infarction, stroke, or cardiovascular death	208 (3.32%)	218 (3.47%)	0.95 (0.79-1.15); p=0.6190	103 (2.72%)	135 (3.45%)	0.79 (0.61-1.02); p=0.0661
Myocardial infarction*	95 (1.52%)	112 (1.78%)	0.85 (0.64-1.11); p=0.2325	37 (0.98%)	72 (1.84%)	0.53 (0.36-0.79); p=0.0014
Non-fatal myocardial infarction	88 (1.40%)	98 (1.56%)	0.90 (0.67-1.20); p=0.4562	32 (0.84%)	60 (1.53%)	0.55 (0.36-0.84); p=0.0056
Stroke*	75 (1.20%)	67 (1.07%)	1.12 (0.80-1.55); p=0.5072	40 (1.06%)	37 (0.95%)	1.12 (0.71-1.75); p=0.6291
Cardiovascular death	38 (0.61%)	39 (0.62%)	0.97 (0.62-1.52); p=0.9010	26 (0.69%)	26 (0.66%)	1.03 (0.60-1.77); p=0.9161
Unstable angina	20 (0.32%)	20 (0.32%)	1.00 (0.54-1.86); p=0.9979	8 (0.21%)	11 (0.28%)	0.75 (0.30-1.87); p=0.5280
Transient ischaemic attack	42 (0.67%)	45 (0.72%)	0.93 (0.61-1.42); p=0.7455	19 (0.50%)	19 (0.49%)	1.03 (0.55-1.95); p=0.9181
Any death	160 (2.55%)	161 (2.57%)	0.99 (0.80-1.24); p=0.9459	108 (2.85%)	101 (2.58%)	1.10 (0.84-1.45); p=0.4796

*Fatal or non-fatal.

Table 2: Efficacy endpoints in the intention-to-treat and per-protocol populations

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Figure 3: Primary outcome by prespecified subgroups (intention-to-treat population)
Hazard ratios are unstratified.



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	Aspirin (n=6270)	Placebo (n=6276)
Total number of serious adverse events	1266 (20.19%)	1311 (20.89%)
Bleeding serious adverse events by severity		
Any gastrointestinal bleed	61 (0.97%)	29 (0.46%)
Severe gastrointestinal bleed	4 (0.06%)	2 (0.03%)
Moderate gastrointestinal bleed	15 (0.24%)	5 (0.08%)
Mild gastrointestinal bleed	42 (0.67%)	22 (0.35%)
Haemorrhagic stroke	8 (0.13%)	11 (0.18%)

Table 3: Serious adverse events in the intention-to-treat population

the placebo group (hazard ratio [HR] 0.96; 95% CI 0.81-1.13; $p=0.6038$). Gastrointestinal bleeding events (mostly mild) occurred in 61 (0.97%) patients in the aspirin group versus 29 (0.46%) in the placebo group (HR 2.11; 95% CI 1.36-3.28; $p=0.0007$). The overall incidence rate of serious adverse events was similar in both treatment groups

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ARRIVE Trial

Sample Size & Study Population

- Enrolled 12,546 persons without cardiovascular disease, high-risk of bleeding, or diabetes to receive aspirin 100 mg daily or placebo

Results

- The occurrence of cardiovascular death, myocardial infarction, unstable angina, stroke, or transient ischemic attack occurred in
 - 269 (4.29%) patients in the aspirin group versus
 - 281 (4.48%) patients in the placebo group
 - HR 0.96; 95% CI 0.81–1.13; $p=0.6038$

Adverse Effect

- Gastrointestinal bleeding events (mostly mild) occurred in
 - 61 (0.97%) patients in the aspirin group versus
 - 29 (0.46%) in the placebo group
 - HR 2.11; 95% CI 1.36–3.28; $p=0.0007$

McNeil JJ, Wolfe R, Woods RL, et al. *Engl J Med*. 2018;379(16):1509–1518.

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ARRIVE Trial: Conclusion

In patients at low-risk for cardiovascular disease, aspirin use resulted in a *significantly higher risk of major hemorrhage* and did not result in a significantly lower risk of cardiovascular disease than placebo.

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Population	Agency & Year	Recommendation	Evidence Grade
Adults aged 40 to 59 years with a $\geq 10\%$ 10-year CVD risk	USPSTF 2022	Individualized decision = evidence indicates that the net benefit of aspirin use is small	C
Adults aged 60 years and above	USPSTF 2022	Do not initiate aspirin for the primary prevention of CVD.	D

Aspirin Use to Prevent Cardiovascular Disease: Preventive Medication
<https://www.uspreventiveservicestaskforce.org/uspstf/recommendation/aspirin-to-prevent-cardiovascular-disease-preventive-medication>

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Newest Data 2025

- **Usage Decline**
 - A January 2025 analysis of NHANES data confirmed a **significant drop in aspirin use among older adults**, reflecting the successful adoption of "de-prescribing" guidelines.
- **Emerging Exceptions**
 - Growing interest (*not yet formal guideline consensus*) in using **coronary artery calcium (CAC) scores** or **Lipoprotein(a) [Lp(a)] levels** to identify the rare cases needing primary prevention
 - For example, a CAC score >100 might tip the scale in favor of aspirin, but this is personalized medicine, not population health.

<https://jamanetwork.com/journals/jama/fullarticle/2829534>
<https://www.tctmd.com/news/aspirin-use-primary-prevention-declining-us-nhanes>

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Question:

Are statins beneficial when used for primary cardiovascular prevention in older adults?

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Population	Agency & Year	Recommendation	Evidence Grade
Adults aged 40 to 75 years with no history of CVD, 1 or more CVD risk factors, and a calculated 10-year CVD event risk of 10% or greater	USPSTF 2016	Initiate low- to moderate-intensity statin if all criteria are met	B
Adults aged 40 to 75 years with no history of CVD, 1 or more CVD risk factors, and a calculated 10-year CVD event risk of 7.5% to 10%	USPSTF 2016	May offer low- to moderate-intensity statin if all criteria are met	C
Adults 76 years and older with no history of CVD	USPSTF 2016	Insufficient evidence	I

Statin Use for the Primary Prevention of Cardiovascular Disease in Adults: Preventive Medicine
<https://www.uspreventiveservicestaskforce.org/uspstf/document/RecommendationStatementFinal/statin-use-in-adults-preventive-medication>

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JAMA Internal Medicine | Original Investigation

Effect of Statin Treatment vs Usual Care on Primary Cardiovascular Prevention Among Older Adults

The ALLHAT-LLT Randomized Clinical Trial

Benjamin H. Han, MD, MPH; David Sutin, MD; Jeff D. Williamson, MD; Barry R. Davis, MD, PhD; Linda B. Piller, MD, MPH; Haninah Pervin, PhD; Sara L. Pressel, MS; Caroline S. Blaum, MD; for the ALLHAT Collaborative Research Group

JAMA Intern Med. 2017;177(7):955-965. doi:10.1001/jamainternmed.2017.1442
 Published online May 22, 2017.

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ALLHAT-LLT Trial

Sample Size & Study Population

- **Post hoc secondary data analyses were conducted of participants 65 years and older without evidence of atherosclerotic cardiovascular disease**
- Pravastatin group: 1467 participants (mean age, 71.3 + 5.2 years)
- Usual care (UC) group: 1400 participants (mean age, 71.2 + 5.2 years)

Outcomes

- Primary outcome: all-cause mortality
- Secondary outcomes
 - cause-specific mortality and nonfatal myocardial infarction or fatal coronary heart disease combined (coronary heart disease events)

Results

- HRs for all-cause mortality in the pravastatin group vs the UC group were
 - 1.18 (95% CI, 0.97-1.42; P = .09) for all adults 65 years and older
 - 1.08 (95% CI, 0.85-1.37; P = .55) for adults aged 65 to 74 years
 - 1.34 (95% CI, 0.98-1.84; P = .07) for adults 75 years and older

Han BH, Sutin D, Williamson JD, et al. JAMA Intern Med. 2017 Jul 1;177(7):955-965.

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ALLHAT-LLT Trial: Conclusion

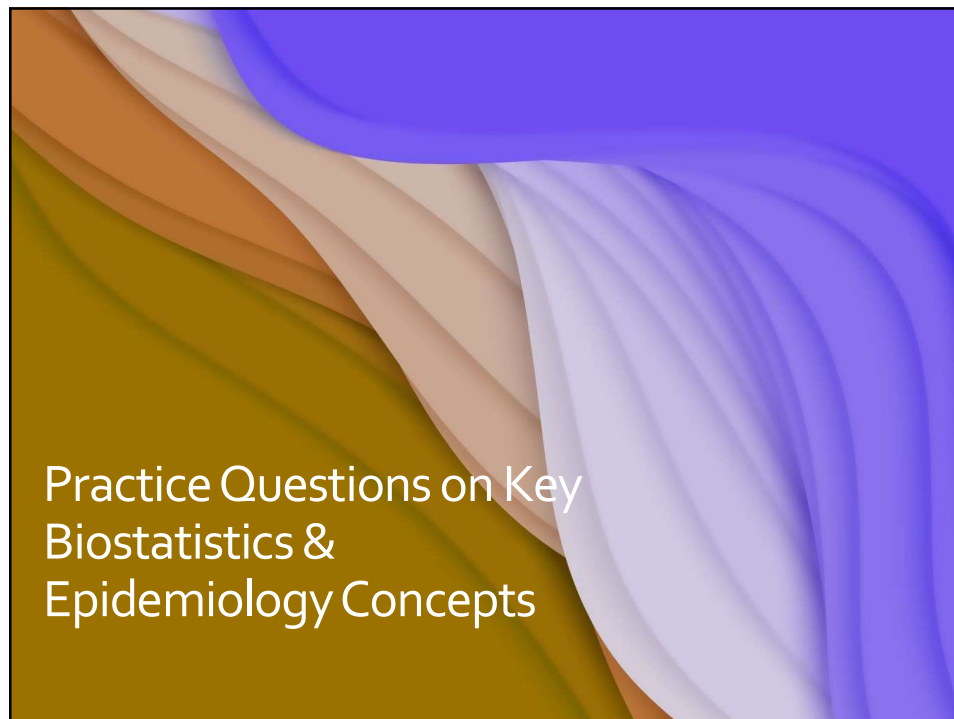
No benefit was found when pravastatin was given for primary prevention to older adults with moderate hyperlipidemia and hypertension, and a nonsignificant direction toward increased all-cause mortality with pravastatin was observed among adults 75 years and older.

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Population	Agency & Year	Recommendation	Evidence Grade
Adults aged 40 to 75 years who have 1 or more CVD risk factors and an estimated 10-year CVD event risk of 10% or greater	USPSTF 2022	Initiate a statin	B
Adults aged 40 to 75 years who have 1 or more CVD risk factors and an estimated 10-year CVD event risk of 7.5% to 10%	USPSTF 2022	May offer low- to moderate-intensity statin selectively	C
Adults 76 years and older with no history of CVD	USPSTF 2022	Insufficient evidence to support for or against	I

Statin Use for the Primary Prevention of Cardiovascular Disease in Adults: Preventive Medication
<https://www.uspreventiveservicestaskforce.org/uspstf/recommendation/statin-use-in-adults-preventive-medication>

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42

Review this table on the incidence of tuberculosis infection among congregated, HIV-infected prison inmates by dormitory wing.

Incidence of Mycobacterium Tuberculosis Infection Among Congregated HIV-Infected Prison Inmates by Dormitory Wing		
	Developed Tuberculosis?	
	Yes	No
East Wing	28	129
West Wing	4	133

A. Calculate the ODDS RATIO of developing tuberculosis infection if residing in the East Wing versus residing in the West Wing and show your work.

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Review this table on the incidence of tuberculosis infection among congregated, HIV-infected prison inmates by dormitory wing.

Incidence of Mycobacterium Tuberculosis Infection Among Congregated HIV-Infected Prison Inmates by Dormitory Wing		
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	Yes	No
East Wing	28	129
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Would you say that your ODDS RATIO is an accurate approximation of the RISK RATIO? Why or why not?

44

A 47-year-old man with low back pain for the past 4 weeks wants to know if you recommend surgery. You prefer to base your treatment recommendations on research evidence whenever possible. In the strongest study you can find, investigators reviewed the medical records of 40 consecutive women with low back pain under care in their clinic – 22 had been referred for surgery, and the other 18 patients had remained under medical care without surgery. The study compared rates of disabling pain after 2 months. All of the surgically treated patients and 10 of the medically treated patients were still being seen in the clinic throughout this time. Rates of pain relief were slightly higher in the surgically treated patients.

For each of the following statements, circle the one response that best represents the corresponding threat to validity.

Because there are relatively few patients in this study, it may give a misleading impression of the actual effectiveness of surgery.

- a. Selection bias
- b. Measurement bias
- c. Confounding
- d. Chance
- e. External validity

The results of this study may not apply to your patient, a man, because all of the patients in this study were women.

- a. Selection bias
- b. Measurement bias
- c. Confounding
- d. Chance
- e. External validity

45

The patients who were referred for surgery were younger and fitter than those who remained under medical care.

- a. Selection bias
- b. Measurement bias
- c. Confounding
- d. Chance
- e. External validity

Compared with patients who had medical care alone, patients who had surgery might have been less likely to report whatever pain they had and the treating physicians might have been less inclined to record pain in the medical record.

- a. Selection bias
- b. Measurement bias
- c. Confounding
- d. Chance
- e. External validity

Patients without other medical conditions were both more likely to recover and more likely to be referred for surgery.

- a. Selection bias
- b. Measurement bias
- c. Confounding
- d. Chance
- e. External validity

46

Bias: What are the following?

Selection Bias	Occurs when <u>comparisons are made between groups of patients that differ</u> in ways other than the one under study
Measurement Bias	Occurs when <u>data are not collected accurately</u> due to issues with measuring device or method
Confounding	Occurs when <u>two factors are associated (travel together)</u> and the effect of one is confused with or distorted by the effect of the other

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Which of the following statements is correct?

- When interpreting relative risks, we look for 1 as the null value, signifying no association between the intervention group and the control group.
- The interpretation between relative risk and odds ratio is same, as the null value for both is 1.
- Odds are defined as the sum of the probability of an event occurring plus the probability of it not occurring.
- Answers A and B above are correct.
- All of the above are correct.

48

The aim of the study was to determine if nebulized epinephrine decreased hospital admission when compared to placebo in patients with a moderate asthma exacerbation treated in the emergency department.

		Admission	
		YES	NO
Treatment	Epinephrine	10	90
	Placebo	80	20

What is the relative risk (RR) of admission in the epinephrine group compared to the placebo group?

- A. $(100/100) / (20/100) = 5$
- B. $(80/100) / (10/100) = 8$
- C. $(10/100) / (90/100) = 0.11$
- D. $(90/100) / (10/100) = 9$
- E. $(10/100) / (80/100) = 0.125$

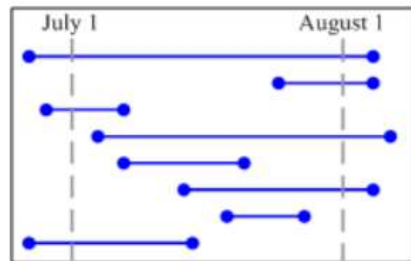
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How would you interpret the relative risk (RR) in the question above?

- a. Patients in the epinephrine group were RR times more likely to admit to the emergency room for treating a moderate asthma exacerbation than patients in the placebo group.
- b. Patients in the epinephrine group were RR times less likely to admit to the emergency room for treating a moderate asthma exacerbation than patients in the placebo group.
- c. Patients in the epinephrine group were RR percentage more likely to admit to the emergency room for treating a moderate asthma exacerbation than patients in the placebo group.
- d. Patients in the epinephrine group were RR percentage less likely to admit to the emergency room for treating a moderate asthma exacerbation than patients in the placebo group.
- e. All of the above are wrong.

50

Assume that the horizontal lines in the diagram represent duration of illness in 8 different people, out of a community of 700.



What is the prevalence of disease during July?

- a. $3 / 700$
- b. $4 / 700$
- c. $5 / 700$
- d. $8 / 700$

What is the incidence of disease during July?

- a. 3
- b. 4
- c. 5
- d. 8

51

The following data is from a large study of bladder cancer and cigarette smoking in Bos

	Bladder Cancer Rates per 100,000 Males
Cigarette smokers	48.0
Nonsmokers	25.4

The relative risk (RR) of developing bladder cancer for male cigarette smokers compared with male nonsmokers is

- a. 48.0
- b. $48.0 - 25.4 = 22.6$
- c. $48.0 / 25.4 = 1.89$
- d. $(48.0 - 25.4) / 48.0$
- e. Cannot be computed from the data given above

The risk difference (RD) of bladder cancer due to cigarette smoking in male cigarette smokers is

- a. $48.0 / 25.4 = 1.89$
- b. $48.0 - 25.4 = 22.6$ per 100,000
- c. 48.0
- d. $48.0 / 100,000 = 0.00048$
- e. Cannot be computed from these data given above

52

Medication A was given to high-risk volunteers for a study to prevent HIV. In the intervention group, 10 developed HIV infection, and 125 didn't. In the control group (no Medication A was given), 47 developed HIV infection, and 259 didn't. What is the odds ratio of developing HIV if received Medication A vs. did not receive Medication A?

- a. 2.27
- b. 0.44
- c. 0.015
- d. 68.9

53

What represents the chance of making a Type I error when conducting research?

- a. 25%
- b. Power
- c. P-value
- d. 5%
- e. Beta

54

A clinician is looking for a good screening test for breast cancer. What is the most important characteristics for a screening test?

- a. High overall test accuracy
- b. High negative predictive value
- c. High positive predictive value
- d. High sensitivity
- e. High specificity

55

A new medication is very effective in patients with a stroke, but deadly in patients who do not have a stroke. What is the most important characteristic for a confirmatory test for stroke, that will ensure as few people without a stroke receive the new medication?

- a. High overall test accuracy
- b. High negative predictive value
- c. High positive predictive value
- d. High sensitivity
- e. High specificity

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What statistical term represents the chance that a research will detect a difference in treatments, if one truly exists?

- a. Power
- b. Beta
- c. Validity
- d. Alpha
- e. P-value

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Questions???

Please do not forget to complete the post-lab assignment.

Will be available at 3:00 pm today.

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